Yash Jain

Senior AI & Backend Engineer

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Relevant Experience

Epam Systems *Bangalore*
Senior AI Engineer *January 2026 – Current*

- Built a **LangGraph-based orchestrator agent** that answers complex business questions by dynamically routing requests to four specialist AI sub-agents covering **Nielsen, Fusion, Growth, and CTA** domains.
- Wrapped specialist agents as **native tools** and implemented LLM-driven tool calling, enabling the planner to generate **custom prompts per sub-agent** instead of forwarding raw user queries.
- Implemented **parallel sub-agent execution** with an observe-and-retry loop and configurable iteration limits, improving multi-agent workflow efficiency and reliability.
- Built a **question-resolution layer** using conversation context to resolve follow-up queries such as *"and in ASEAN?"*, enabling accurate multi-turn routing and entity resolution.
- Managed **LangGraph state, reducers, and checkpointing** with per-agent conversation threads and windowed chat history, enabling persistent multi-turn context across agent interactions.
- Designed **conditional routing** to dynamically direct requests across context-answer, domain-query, and out-of-scope paths, while merging clarification requests from multiple agents into a single coherent response.
- Implemented a **two-tier LLM architecture** using a high-capability model for final synthesis and a low-latency model for routing, with **multi-endpoint fallbacks** to improve system availability.
- Built **end-to-end turn tracing and benchmark evaluations** to debug agent routing, prompts, and entity resolution, validating multi-turn conversation quality across representative business queries.

Quantzig | AB InBev *Bangalore*
Senior AI Engineer *April 2024 – January 2026*

- Developed an **LLM-based Vendor Statement Reconciliation** system using **FastAPI, LangChain, and LangGraph**, building a complete StateGraph AI workflow that **reduced manual reconciliation effort by ∼70%**.
- Automated detection of Overpayments, Underpayments, Missing Invoices, and Obsolete transactions using **Python and pandas**, improving **financial reporting accuracy by ∼40%**.
- Implemented **LLM-based OCR** to extract invoice data from SAP PDFs, cutting **data extraction time by ∼60%** and ensuring precise validation of over/underpayments.
- Integrated **LangSmith** monitoring across all LLM pipelines, improving **workflow observability and reducing debugging time by ∼35%**.
- Led **end-to-end project delivery** across requirement gathering, architecture, development, and testing with **GitHub Actions CI/CD** across multiple regions, **reducing deployment time by ∼50%**.
- Built a **Duplicate Payment Detection** system using **Azure Data Factory, Azure Blob Storage**, and fuzzy matching logic, **flagging ∼95% of duplicate transactions** before processing.

Evabot *Remote*
Senior AI & Backend Engineer *Oct 2023 – Apr 2024*
AI & Backend Engineer *Jun 2022 – Sep 2023*

- Architected and owned the **end-to-end ML backend platform**, improving **system latency by ∼30%** using **FastAPI, Celery, and MongoDB**.
- Built a **hybrid RAG retrieval system** (sparse + dense vectors with **Pinecone**), improving **document retrieval accuracy by ∼35%** over single-vector search.
- Implemented **streaming LLM responses** (GPT-4 / GPT-4 Turbo), reducing **perceived response latency by ∼40%** for end users.
- Designed and launched **Rapport Dashboard** and **Sales Enablement** extension, boosting **sales productivity by ∼25%** with real-time AI insights via Pusher.
- Engineered a **multithreaded batch processing pipeline** with **Celery workers**, reducing **Vector DB ingestion time by ∼45%** while maintaining safe CPU utilization.
- Developed a **large-scale web crawler** for user and company enrichment, reducing **manual research effort by ∼80%**.

- Built a **Nudge Generation system** using **Selenium-based crawlers**, reducing **data preparation time by ∼65%**.
- Implemented **Redis caching** for high-frequency APIs, improving **average response times by ∼50%**.
- Used **GPT-4** for personalized email generation and sales nudges, improving **email engagement rates by ∼30%**.
- Built a **hybrid recommendation engine** (content-based + collaborative filtering), improving **recommendation CTR by ∼20%**.
- Containerized services using **Docker & docker-compose** on **AWS EC2**, maintaining **99.5% production uptime**.

Dell *Bangalore*

Software Developer Engineer 1 *Sep 2021 – June 2022*

- Engineered a **scalable Recommendation System** with a **batch-processing framework** capable of handling **millions of requests per day** with low latency.
- Integrated recommendations into **Dell's global e-commerce platform**, contributing to a **∼15% increase in accessory and warranty upsell revenue**.
- Deployed **online and offline recommendation modes** — personalized website suggestions and email campaigns — **boosting user engagement by ∼18%**.

Smart Controls *Bangalore*

Software Developer Engineer *Dec 2019 – Sep 2021*

- Developed **IIoT predictive applications** for tyre production forecasting, enabling **∼20% better resource planning** through daily inventory and availability modeling.
- Transitioned from **Linear/Ridge Regression to Random Forest**, achieving a **∼30% improvement in prediction accuracy** on high-redundancy industrial datasets.
- Integrated predictive models into a **Flask API** with optimized data pipelines, reducing **real-time inference latency by ∼25%**.

Agrogram *Remote*

Backend Developer Intern *May 2019 – June 2019*

- Built **RESTful APIs** using **Node.js, Express, and Mongoose ORM** for a farming e-commerce Android app, **reducing average API response time by ∼20%** through optimized query design.

SKILLS

Programming Tools: Python, FastAPI, Flask, Flask Restful, Celery, Django, Javascript, Nodejs, Express, Docker, Git, Rest API, Selenium, Beautiful Soup, Microservice Architecture, Azure Data Factory (ADF)

Database: MongoDB, SQL, Pinecone (Vector DB)

AI & ML: OpenAI Models & API, Claude (Sonnet, Haiku, Opus), Hugging Face, LangChain, LangGraph, Model Context Protocol (MCP), LlamaIndex, OpenAI Agent SDK, Claude Agent SDK, Pydantic AI, Microsoft Agent Framework, ReAct Agents, LangSmith, Scikit-learn, NumPy, Pandas, Matplotlib, PySpark, Azure Databricks, Ollama, OpenWebUI, Claude Code

PROJECTS

Curated Gift Boxes Recommendation System

- Designed a **profit-optimized recommendation engine** for gift boxes using user preference data, **improving profit margins by ∼12%** through smart bundling strategies.

IMDB Sentiment Analysis Web App

- Built a **full-stack sentiment analysis app** on IMDB reviews using **NLP models and Selenium**, achieving **∼91% classification accuracy** on the test dataset.

Quora Duplicate Question Detection (NLP)

- Trained an **NLP classification model** on the Kaggle Quora dataset, achieving **∼88% F1-score** for detecting semantically duplicate question pairs.

Flipkart Web Scraper

- Developed a **robust product data scraper** for Flipkart using **BeautifulSoup and Selenium**, capable of extracting **10,000+ product records per hour** reliably.

Academic Qualification

Madhav Institute of Technology & Science Gwalior – RGPV University
B.E. (Computer Science) | CGPA: 7.85 *2016 – 2020*

The Radiant Higher Secondary School – CBSE
12th | 82.2% *2014 – 2015*

Rewards & Recognition

- Earned certification in **Model Context Protocol: Advanced Topics**.
- Earned certification in **Introduction to Agent Skills**.
- Earned certification in **Introduction to Model Context Protocol**.
- Earned certification in **Claude Code in Action**.
- Earned certification in **Claude 101**.
- **AIR 70** in TCS CodeVita Season 8 – Competitive Programming Competition (Zonal Rank: 49).
- Scored **98.61 percentile** in Quantitative Aptitude in AMCAT.
- **Rank 2591** in CodeChef SnackDown 2017 Pre-Elimination Round.